# ZeGA: Zero-Gradient Gated Adaptation via Weight-Space Synapse Cartridges

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**Code & Benchmarks:** https://github.com/Aphygo/zega-synapse-cartridges

**Cartridges:** hf.co/Aphygo/zega-ministral3-8b-swedish-v1 · hf.co/Aphygo/zega-clip-vit-b32-polarization-v1

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**Abstract**

**ZeGA** (**Zero-Gradient Gated Adaptation**) compiles input-selective residual branches into appended Transformer MLP channels **without gradient optimization**. Building on the *Weight-Space Programming* paradigm [24], we evaluate reversible sub-megabyte `.safetensors` **Synapse Cartridges** (`aphygo.synapse_cartridge.v1`) in a language model, a vision–language model, and an SDXL-derived image generator. In the adaptation regimes studied here, gradient-trained low-rank adapters (`LoRA`, `DoRA`) [11, 15] require iterative backpropagation and apply shared weight deltas across positions; locate-and-edit factual editors (`ROME`, `MEMIT`, `AlphaEdit`) [17, 18, 5] optimize target vectors via gradient descent prior to overwriting base `down_proj` weights in-place; and constant post-norm activation steering [33] shifts opposing poles unconditionally on symmetric contrasts.

ZeGA synthesizes closed-form **structural width expansions** ($M \to M + K_\ell$) from forward-pass calibration activations and attaches them directly to native `nn.Linear` layers via a single 272-line zero-wrapper PyTorch loader (`cartridge_loader.py`), leaving 100% of preexisting base channel weights untouched. Each appended channel contributes a full-dimensional ($\mathbb{R}^d$) output vector (per-block span rank $\leq K_\ell$) modulated by a nonlinear gate–value product (**Family I**: `SwiGLU`, `GeGLU`, `ReGLU`) or an analytically constructed polarization group (**Family II**: standard non-gated 2-layer MLPs with `GELU`, `QuickGELU`, or `SiLU`), for which we prove (**Theorem 3**) closed-form 2-row stationary-minimum and 4-row fixed-center symmetric quartet identities whose leading local Taylor term isolates the bilinear cross-product $g\,u$ without a `gate_proj` matrix (with **Family III** convolutional bottlenecks formulated as a mathematical extension in **Corollary 1**):

1. **Language (`Ministral-3-8B-Instruct-2512-BF16` [19], rev. `f6fae979`, Family I `SwiGLU`,** 960.95 **KiB):** Appending +8/14,336 channels in L32 and +32/14,336 in L33 improves `SwedMaster-300` from 73.67% to 84.00% (+10.33 pp, $p = 1.64 \times 10^{-6}$), `SwedFreeGen-60` from 70.00% to 95.00% ($p = 2.75 \times 10^{-4}$), and the first 1,000 test items of `ScaLA-sv` [20] from 71.30% to 75.40% ($p = 0.00864$), while preserving `Belebele-SV/EN/FR` [1] (300/200/200 items: 88.0%/94.0%/90.0%, 0 per-item flag flips), a `DaLAJ`-derived minimal-pair evaluation [27] ($n = 500$: 90.80% → 90.00%, $p = 0.3438$), `Anchor-40` (40/40 Top-1 match, end-to-end $\overline{D}_{\mathrm{KL}} = 5.66 \times 10^{-5}$), and character-for-character identical greedy completions on both `Python-20` (20/20 pass, 20/20 char-identical) and `Math-20` (7/20 → 7/20, 20/20 char-identical), with 0 calibration string overlaps across all 2,640 items and 100.0000% full-byte SHA-256 restoration across all 6 modified `SwiGLU` matrices and full `lm_head` (8.40 ms hot-swap).
2. **Vision–Language (`openai/clip-vit-base-patch32` [22], rev. `3d74acf9`, Family II `QuickGELU`,** 336.80 **KiB):** Compiled in 26.6 s without gradient optimization (+56/3,072 channels in visual L10) from single-object renders, same-color centering scenes (0 cross-color multi-object scenes seen), disjoint natural anchors [13, 16], and strictly template-disjoint

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calibration descriptors (0/574 caption or template overlap with any evaluation suite), our 4-row quartet (**Theorem 3b**) raises Winoground-style Pair Group Score [26, 30] across 290 completely distinct unseen-seed rendered image pairs (580 unique images): on `CompBind-240` (120 image pairs) from 0.83% to 25.83% (31 wins vs. 1 loss, within-suite $p = 1.54 \times 10^{-8}$; I2T 52.92% $\rightarrow$ 89.17%, $p = 5.11 \times 10^{-20}$), on clause-inverted `ClauseSwap-180` (90 independent image pairs) from 2.22% to 34.44% (30 wins vs. 1 loss, $p = 2.98 \times 10^{-8}$; I2T 52.78% $\rightarrow$ 81.67%, $p = 1.81 \times 10^{-9}$), and on `SpatialRel-160` (80 image pairs) from 0.00% to 6.25% (5 wins vs. 0 losses, Group $p = 0.0625$, below the 16.67% i.i.d. continuous-score reference while lifting I2T from 50.62% to 65.00%, $p = 0.00140$), whereas a 1-row control stays at 0.83%/3.33%/0.00% Group ($p = 1.00$). On 500 `CIFAR-100` test photos and 200 held-out `CelebA` portraits (`huggan/CelebA-faces-with-attributes` rev. `b47e27a7`), the quartet preserves 65.80% zero-shot accuracy (499/500 Top-1 agreement, $\overline{D}_{\mathrm{KL}} = 1.99 \times 10^{-9}$; 2-row pair 500/500 at $9.74 \times 10^{-8}$) and 1.0000000 embedding cosine (200/200 probe agreement; 0.40 ms hot-swap).

3. **Generative Image Diffusion (`Juggernaut XL`, SDXL-derived [21] `UNet2DConditionModel`, `GEGLU`, 906.84 KiB):** Appending +8/5,120 channels across 15 $d = 1280$ `GEGLU` blocks (compiled from 4 single-object vessel pairs and 5 anchors, with scalar offset/gain tuned on 1 development scene `s104`), our patch-gated cartridge converts porcelain vessels into sapphire and gold filigree (0.16% $\rightarrow$ 85.94% across all 8 multi-object scenes; 83.99% excluding exact development scene `s104`, $n = 7$; 81.33% across the 6 scenes from the 3 other prompt families) while preserving nearby bystanders (85.91% all-8, 83.90% $n = 7$, 81.21% $n = 6$ naturalness vs. 4.11%/4.70%/5.09% for a matched constant-gate control; binary 55%-quantile off-target proxy PSNR 16.36 vs. 10.35 dB all-8, 16.15 vs. 10.22 dB $n = 6$). On 4 target-free anchors, 100.00% of cells on the signed mean positive-CFG-branch `GELU` gate map (without value product) stay $\leq 0.01$, yielding 32.31 dB full-image PSNR (0.9495 SSIM) vs. 10.22 dB (0.2297 SSIM).

# 1 Introduction

Feed-forward **Expand-Activate-Contract** channel blocks—implemented as gated linear units (`SwiGLU`, `GeGLU`, `ReGLU`) [23] or standard two-layer key-value MLPs (`GELU`, `QuickGELU`, `SiLU`) [9, 7]—provide wide intermediate channel banks in language ($M = 14{,}336$ in `Ministral-3-8B`), vision–language ($M = 3{,}072$ in `CLIP-ViT-B/32`), and SDXL diffusion backbones ($M = 5{,}120$ in $d = 1280$ `GEGLU` blocks) [21]. Recently, Takehi, Teoh, A. Zhang, and Khattab [24] introduced *Weight-Space Programming (WSP)*, directly synthesizing neural weights without gradient optimization. Extending WSP from compact circuits and single-direction loop dampening (*Sparks* Experiment 8) to reversible, context-selective adaptation in foundation models requires addressing three concrete challenges: (i) constant post-norm steering shifts opposing poles symmetrically while spatially uniform diffusion adapters alter bystanders; (ii) Vision Transformers (`CLIP-ViT` [22]) lack multiplicative `gate_proj` matrices; and (iii) subword BPE tokenizers split inflected words across shared stems while upstream lookahead cues must traverse blocks with low intermediate readout drift.

**ZeGA** compiles narrow width expansions ($M \rightarrow M + K_\ell$) from forward-pass calibration activations without gradient optimization into detachable `.safetensors` **Synapse Cartridges** (`aphygo.synapse_cartridge.v1`), concatenated onto native `nn.Linear` layers via a 272-line loader (`cartridge_loader.py`) with zero runtime hooks.

# 2 Related Work

**Width Expansion & Neuron Patching.** *Net2WiderNet* [3] replicates channels and rescales outgoing weights to preserve function before SGD. *CaliNet* [4] adds a trainable parallel calibration FFN, and *Transformer-Patcher* [12] trains input-selective key/bias/value neuron patches in the final FFN via gradient descent. ZeGA shares width expansion ($M \rightarrow M + K_\ell$), but synthesizes weights *without gradient optimization*, proves explicit second-order polarization identities (**Theorem 3**) for non-gated MLPs, assembles native `SwiGLU`/`GEGLU` gate–value products, and validates cartridges across language, `CLIP-ViT`, and `SDXL`.

Table 1: Comparison of post-training adaptation mechanisms.

| Method | Optimization / Fitting | Base Weights | Local Update | Input Selectivity | Runtime |
|---|---|---|---|---|---|
| `LoRA` [11] | Gradient descent | Frozen; merge writes $\mathbf{W}_0 + \mathbf{BA}$ to target matrices | Linear $\Delta\mathbf{W} = \mathbf{BA}$ (rank $\leq r$) | Linear $\mathbf{BAx}$ | Native (merged) or adapter |
| `DoRA` [15] | Gradient descent on mag. + LoRA dir. | Frozen; merge overwrites target matrices | Renormalized cols. (rank can $> r$) | Input-independent $\Delta\mathbf{W}$ | Native (merged) or adapter |
| `CaliNet` / `T-Patcher` [4, 12] | Gradient descent on $+K$ FFN slots | 0% base mutated | Span $\leq K$ in $\mathbb{R}^d$ | Gradient-trained input-selective neuron patch | Native text MLP |
| `ROME` / `MEMIT` / `AlphaEdit` [17, 18, 5] | Analytic weight update after gradient target opt. | Overwrites `down_proj` in-place | Rank-1 (`ROME`) or rank $\geq 1$ (`MEMIT`, `AlphaEdit`) | Entity key match without conjunctive gate | Mutated `nn.Linear` |
| Const. Post-Norm Steering [33] | Reading vector (0 grad steps) | 0% | Rank-1 $+\alpha\mathbf{v}$ | None (constant shift) | Residual hook |
| **ZeGA (Ours)** | **Closed-form SVD / Ridge / Polarization (0 grad)** | 0% ($+K_\ell$ **detachable**) | $\mathbb{R}^d$ **cols. (span** $\leq K_\ell$**)** | **Gate–value product or quartet (Thm 3)** | **Native `nn.Linear` (272-line loader)** |

**Model Editing, Routing Memories & Nullspace PEFT.** *ROME* [17], *MEMIT* [18], and *AlphaEdit* [5] optimize target representations via Adam backpropagation before solving normal equations to overwrite `down_proj` (with *AlphaEdit* projecting onto preserved input-key nullspaces), whereas ZeGA leaves base weights intact and gates every appended channel. External routing editors (*GRACE* [8], *WISE* [28]) introduce deferral codebooks or trained side FFNs, whereas ZeGA uses native `nn.Linear` GEMMs. In PEFT (*LoRA* [11], *DoRA* [15]), *LoRA-Null* [25] initializes factors via SVD in calibrated activation nullspaces ($\mathbf{W}_0 - \mathbf{BA}$) prior to SGD, while *Null-LoRA* [32] cross-freezes factors and projects trained increments onto $\ker(\mathbf{W}_0^\top)$. *Representation Engineering* [33] includes both trained `LoRRA` adapters and constant reading-vector shifts (analyzed in Proposition 1).

**Diffusion Locality & Weight-Space Programming.** *Concept Sliders* [6] train low-rank adapters via contrastive denoising; our Constant-Gate Control specifically ablates ZeGA's spatial gate at fixed $(\mathbf{U}_{\text{ext}}, \mathbf{b}_u, \mathbf{D}_{\text{ext}})$. *LoRA-Composer* [29] composes pre-trained LoRAs using attention layout constraints and inference-time latent gradient updates, while *RegionRoute* [2] trains a mask-supervised LoRA-MoE on Flux.1-Kontext for mask-free inference. ZeGA instead compiles spatial gates directly into `UNet GEGLU` projections without gradient optimization or masks. Relative to *Sparks of Weight-Space Programming* [24] (including `Qwen3.5-0.8B` loop dampening in Experiment 8) and *WeightLoop* [14] (where a fixed editor updates assay/gene-ranking predictors from experimental feedback), ZeGA establishes multi-channel gated cartridges across language, vision–language, and diffusion.

# 3 Mathematical Formulation of ZeGA Synapse Cartridges

## 3.1 Expand-Activate-Contract Width Expansion ($M \to M + K_\ell$)

In **Family I (Gated MLPs: `SwiGLU`, `GeGLU`, `ReGLU`)**, appending $K_\ell$ rows $\mathbf{G}_{\text{ext}}^{(\ell)}, \mathbf{U}_{\text{ext}}^{(\ell)} \in \mathbb{R}^{K_\ell \times d}$ (with optional $\mathbf{b}_g, \mathbf{b}_u \in \mathbb{R}^{K_\ell}$) and columns $\mathbf{D}_{\text{ext}}^{(\ell)} \in \mathbb{R}^{d \times K_\ell}$ yields:

$$\text{MLP}_\ell^{\text{ZeGA}}(\mathbf{x}) = \underbrace{\text{MLP}_\ell^{\text{orig}}(\mathbf{x})}_{\text{Unmodified Base Channels } (1\ldots M)} + \sum_{k=1}^{K_\ell} \underbrace{\left[\sigma_{\text{act}}(\mathbf{g}_k^\top \mathbf{x} + b_{g,k}) \cdot (\mathbf{u}_k^\top \mathbf{x} + b_{u,k})\right]}_{\alpha_k(\mathbf{x}) \in \mathbb{R} \text{ (Nonlinear Gate–Value Product)}} \mathbf{d}_k. \tag{1}$$

For fused chunked `GEGLU` blocks (`SDXL ff.net[0].proj`: $\mathbb{R}^d \to \mathbb{R}^{2M}$), `cartridge_loader.py` interleaves $\mathbf{W}_{\text{proj}}^{\text{new}} = [\mathbf{W}_{\text{up}}; \mathbf{U}_{\text{ext}}; \mathbf{W}_{\text{gate}}; \mathbf{G}_{\text{ext}}]$ and $\mathbf{b}_{\text{proj}}^{\text{new}} = [\mathbf{b}_{\text{up}}; \mathbf{b}_u; \mathbf{b}_{\text{gate}}; \mathbf{b}_g]$, matching Eq. (1).

### 3.2 Linear Margin Trade-Off & Local First-Branch BPE Gating

**Proposition 1** (Linear Margin Trade-Off Under Constant Post-Norm Shifts)**.** *At a fixed bias-free linear readout* $\Delta\mathbf{w} = \mathbf{w}_{w_+} - \mathbf{w}_{w_-}$ *over opposing context sets* $\mathcal{C}_+$ *and* $\mathcal{C}_-$*, any constant post-norm shift* $\mathbf{h}(\mathbf{x}) = \mathbf{h}_0(\mathbf{x}) + \mathbf{v}$ *with* $\delta = \Delta\mathbf{w}^\top\mathbf{v} > 0$ *flips every* $\mathbf{x} \in \mathcal{C}_-$ *satisfying* $0 < (\mathbf{w}_{w_-} - \mathbf{w}_{w_+})^\top\mathbf{h}_0(\mathbf{x}) < \delta$*. Under* `RMSNorm` *[31]* $\mathrm{rms}_\varepsilon(\mathbf{z}) = \sqrt{\frac{1}{d}\|\mathbf{z}\|_2^2 + \varepsilon}$ *on a direct bias-free readout, a pre-norm shift* $\mathbf{h}_{pre} + \tilde{\mathbf{v}}$ *with* $s' = \mathrm{rms}_\varepsilon(\mathbf{h}_{pre} + \tilde{\mathbf{v}}) > 0$ *preserves the sign of the* new readout contrast $(m_{num} + \delta_{num})/s'$*, whereas the margin change vs. baseline* $(s = \mathrm{rms}_\varepsilon(\mathbf{h}_{pre}))$ *is* $\delta_{num}/s' + m_{num}(1/s' - 1/s)$*, whose sign may differ from* $\delta_{num}$ *when* $s' \neq s$*.*

**Proposition 2** (Local First-Branch Discrimination Under BPE Tokenization)**.** *Fix an identically tokenized prompt prefix* $\mathbf{c}$ *under which* $(w_+, w_-)$ *tokenize as* $[t_0^+, \ldots, t_{m_+-1}^+]$ *and* $[t_0^-, \ldots, t_{m_--1}^-]$ *with first diverging index* $j^* = \min\{j \geq 0 : t_j^+ \neq t_j^-\}$ *(for strict prefixes* $m_- = j^* < m_+$*, completion of* $w_-$ *is defined over an explicit set* $\mathcal{B}$ *of word-boundary continuations disjoint from* $t_{j^*}^+$*). For all* $j < j^*$*,* $t_j^+ = t_j^-$ *contributes an identical factor to both sequences while perturbing unrelated words sharing* $[t_0^+, \ldots, t_j^+]$*;* $j^*$ *is the* earliest step where a single-step local logit contrast directly differentiates $w_+$ from $w_-$ *(full sequence log-ratios also include steps* $j > j^*$*). Verified in-context in* `Ministral-3-8B` *Tekken BPE (*`pillar1_full_6matrix_and_2640_manifest.json`*): prefix* `"Det var ett mycket"` $= [7181, 2841, 13528, 79699]$ + `" politiskt"` $= [3998, 81910]$ *vs.* `" politisk"` $= [3998, 3636]$ *(*$j^* = 1$*); prefix* `"Utredningen saknade ett"` + `" egentligt"` $= [1324, 8029, 38993]$ *vs.* `" egentlig"` $= [1324, 8029, 7249]$ *(*$j^* = 2$*); and prefix* `"Barnen stannade kvar"` + `" hemma"` $= [11146, 1831]$ *vs.* `" hem"` $= [11146]$ *followed by* $\mathcal{B}$ *(*$j^* = 1$*).*

### 3.3 Family II Polarization Identities for Non-Gated MLPs

In non-gated 2-layer MLPs (**Family II**: $\mathbf{W}_{\text{down}}\phi(\mathbf{W}_{\text{up}}\mathbf{x} + \mathbf{b}_{\text{up}})$), a 1-row control $\phi(c + g + u)\mathbf{d}$ contaminates $gu$ with $g^2 + u^2$, $\phi(c)$, and (whenever $\phi'(c) \neq 0$) linear $\phi'(c)(g + u)$.

**Theorem 3** (Stationary-Minimum & Symmetric Quartet Polarization for Family II MLPs)**.** *Let* $\phi(z) = z\sigma(\beta z)$ *(*`SiLU` *at* $\beta = 1$*,* `QuickGELU` *at* $\beta = 1.702$*) or Gaussian* `GELU` $\phi(z) = z\Phi(z)$ *[9], with* $g = \mathbf{g}_k^\top\mathbf{x}$*,* $u = \mathbf{u}_k^\top\mathbf{x}$*, and centering scalar* $c = c(\mathbf{x}) \in \mathbb{R}$*:*

1. ***(Theorem 3a: 2-Row Stationary-Minimum Pair.)*** *Each* $\phi$ *has a unique global minimum* $\tau_0^* < 0$ *with* $\phi'(\tau_0^*) = 0, \phi''(\tau_0^*) > 0$*: analytical values* $(\tau_0^*, \phi(\tau_0^*), \phi''(\tau_0^*)) \approx (-0.751154, -0.163610, 0.370716)$ *for* `QuickGELU`*,* $(-0.751792, -0.169971, 0.431494)$ *for Gaussian* `GELU`*, and* $(-1.278465, -0.278465, 0.217812)$ *for* `SiLU`*. With* $c(\mathbf{x}) = \tau_0^* + e(\mathbf{x})$*, in real arithmetic:*

$$\frac{1}{2\phi''(\tau_0^*)}\Big[\phi(c(\mathbf{x}) + g + u) - \phi(c(\mathbf{x}) + g - u)\Big]\mathbf{d}_k = (g\,u + e(\mathbf{x})\,u)\mathbf{d}_k + \mathcal{O}\Big(\big(|e(\mathbf{x})| + |g| + |u|\big)^3\Big)\mathbf{d}_k. \tag{2}$$

2. ***(Theorem 3b: 4-Row Fixed-Center Symmetric Quartet.)*** *For* every fixed $c \in \mathbb{R}$ *in real arithmetic,*

$$\mathcal{Q}_4(c, g, u) = \Big[\phi(c + g + u) + \phi(c - g - u)\Big] - \Big[\phi(c + g - u) + \phi(c - g + u)\Big] \tag{3}$$

*satisfies* $\mathcal{Q}_4(c, -g, u) = -\mathcal{Q}_4(c, g, u) = \mathcal{Q}_4(c, g, -u)$*, vanishes identically when* $g = 0$ *or* $u = 0$*, and expands locally as* $4\phi''(c)\,g\,u + \mathcal{O}(\|(g, u)\|^4)$ *(and* $\mathcal{Q}_4(0, g, u) = 2\psi_{even}(g + u) - 2\psi_{even}(g - u)$ *at* $c = 0$*). For* $z > 0$ *in real arithmetic, negative tails decay as* $|\phi(-z)| = z/(1 + e^{\beta z})$ *(*`SiLU`*/*`QuickGELU`*) and* $|\phi(-z)| = z\Phi(-z)$ *(Gaussian* `GELU`*).*

### 3.4 Local Spatiotemporal Patch-Gated Bound & Stealth Carrier Sketch

**Corollary 1** (Local Spatiotemporal Patch-Gated Residual Bound & Family III Extension)**.** *In spatial* `GEGLU`*/*`SwiGLU` *blocks (Family I) or proposed convolutional bottlenecks* $C_{mid} \to C_{mid} + K_\ell$ *(Family III),*

*if* $|\sigma_{act}(\mathbf{g}_k^\top \mathbf{x}_{h,w,t} + b_{g,k})| \le \varepsilon_k$ *and* $|\mathbf{u}_k^\top \mathbf{x}_{h,w,t} + b_{u,k}| \le U_k$*, then* $\|\Delta\mathbf{h}^{(\ell)}_{h,w,t}\|_2 \le \sum_{k=1}^{K_\ell} \varepsilon_k U_k \|\mathbf{d}_k\|_2$*. In real arithmetic, smooth gates are strictly non-zero at finite negative arguments (*$|\mathrm{GELU}(-3.852)| \approx 2.26 \times 10^{-4}$*), whereas* `bf16` *values depend on kernel rounding and underflow; global pixel preservation must be audited via PSNR and SSIM.*

For multi-layer lookahead from block $\ell - 1$ to $\ell$, upstream Stealth Context Carriers write rank-1 updates:

$$\Delta\mathbf{h}^{(\ell-1)}_{\text{carrier}}(\mathbf{x}) = (k_{\text{null}} \otimes \delta_{\text{target}})(\mathbf{x}) = k_{\text{null}}\, \delta_{\text{target}}(\mathbf{x}). \tag{4}$$

In our compiler sketch, $k_{\text{null}}$ is orthogonalized against $\le 64$ right singular directions of the first 2,048 centered $\boldsymbol{\gamma}_{\text{norm}}$-weighted unembedding rows and $\le 64$ `Z_avoid` directions; this sketch gives no quantitative bound over all vocabulary rows of $\mathbf{A}k_{\text{null}}$ ($\mathbf{A} = \mathbf{W}_{\text{out}}\,\mathrm{diag}(\boldsymbol{\gamma}_{\text{norm}})$). As an *idealized condition*, for a direct bias-free readout with exact $\mathbf{A}k_{\text{null}} = \mathbf{0}$, positive `RMSNorm` rescaling $\mathrm{rms}_\varepsilon(\mathbf{h})/\,\mathrm{rms}_\varepsilon(\mathbf{h} + ak_{\text{null}}) > 0$ preserves greedy `argmax` (under `LayerNorm` centering $\mathbf{C}$, the exact numerator condition is $\mathbf{AC}k_{\text{null}} = \mathbf{0}$, for which $\mathbf{1}^\top k_{\text{null}} = 0$ with $\mathbf{A}k_{\text{null}} = \mathbf{0}$ is sufficient). End-to-end drift of the full cartridge is measured via `Anchor-40` KL.

# 4 Pillar I: Language Validation on `Ministral-3-8B-Instruct-2512-BF16`

On `mistralai/Ministral-3-8B-Instruct-2512-BF16` [19] (commit `f6fae979`, 8,918,026,240 params), we append 40 `SwiGLU` channels (+8/14,336 in L32, +32/14,336 in L33; 960.95 KiB, 8.40 ms ON / 2.55 ms OFF). Programmatic string auditing (`verify_pillar1_full_6matrix_and_2640_manifest.py`) confirms 0 exact string overlaps between all 185 calibration strings and *all* 2,640 *evaluation items* across all 10 suites (`SwedMaster-300`, `SwedFreeGen-60`, `Anchor-40`, `Python-20`, `Math-20`, `ScaLA-sv` [20] first 1,000 of 2,048 test items, `DaLAJ`-derived minimal pairs $n = 500$ from SuperLim v2 test [27], and `Belebele` [1] first 300/200/200 4-choice items). Full-byte SHA-256 digests match 100.0000% across *all* 6 *modified `SwiGLU` matrices* (`L32.{gate,up,down}_proj`, `L33.{gate,up,down}_proj`) and the *full `lm_head.weight`* before attach, across all 14,336 unmodified base slices while attached, and after `detach()` (Layers 0–31 share identical `data_ptr()`).

Table 2: Paired evaluation on `Ministral-3-8B-Instruct-2512-BF16` (`before_after_verified_results.json`).

| **Benchmark / Audit Suite** | $n$ | **Base** | **`+ZeGA Cartridge`** | **Delta / Exact McNemar $p$** |
|---|---|---|---|---|
| `SwedMaster-300` (All 6 Categories) | 300 | 73.67% | **84.00%** | **+10.33** pp (37W/6L, $p = 1.64 \times 10^{-6}$) |
| — `C1` (*de/dem*) / `C4` (Colloc.) / `C6` (V2/BIFF) | $50 \times 3$ | 62%/72%/98% | **88%**/**88%**/**98%** | `C1`: $p = 2.44 \times 10^{-4}$; `C6`: 49/50 → 49/50 |
| `Symmetric-150` (Const. Steering = 54.67%) | 150 | 59.33% | **72.67%** | +13.33 pp (Empirical illustration of Prop. 1) |
| `SwedFreeGen-60` (Defect-Free Generation) | 60 | 70.00% | **95.00%** | **+25.00** pp (16W/1L, $p = 2.75 \times 10^{-4}$) |
| `ScaLA-sv` (First 1,000 of Test Split) | 1,000 | 71.30% | **75.40%** | **+4.10** pp (137W/96L, $p = 0.00864$) |
| `DaLAJ`-Derived Minimal Pairs (`SuperLim v2`) | 500 | 90.80% | **90.00%** | −0.80 pp (3W/7L, $p = 0.3438$) |
| `Belebele-SV / EN / FR` (First 300/200/200) | 700 | 88.0/94.0/90.0% | **88.0**/**94.0**/**90.0%** | 0 per-item flag flips ($p = 1.0000$) |
| `Anchor-40` / `Python-20` / `Math-20` | 80 | $40/40 \cdot 20/20 \cdot 7/20$ | **40/40 · 20/20 · 7/20** | 40/40 char-id. Code+Math; $\overline{D}_{\text{KL}} = 5.66 \times 10^{-5}$ |

# 5 Pillar II: Non-Gated Vision–Language Validation on `CLIP-ViT-B/32`

On `openai/clip-vit-base-patch32` [22] (commit `3d74acf9`, 151,277,313 params; 100% full-model SHA-256 match after `detach()`), Layer 10 polarization cartridges were compiled in 26.56 s without gradient optimization from 216 single-object renders, 36 same-color renders (0 cross-color multi-object scenes), disjoint `CIFAR-100` [13] (`aadb3af7` train vs. 500 test) and `CelebA` [16] (`huggan/CelebA-faces-with-attributes` rev. `b47e27a7`, `[0:80]` vs. `[80:280]`) anchors, and 840 synthetic calibration descriptors having 0/574 (0.00%) **string or template overlap** with any evaluation caption. Pixel SHA-256 auditing confirms 290/290 **completely distinct unseen-seed**

**rendered image pairs (580 unique images)** across `CompBind-240` (120 pairs, seeds $\geq$ 5000), `ClauseSwap-180` (90 independent pairs, seeds $\geq$ 6500), and `SpatialRel-160` (80 pairs, seeds $\geq$ 8000; within-suite McNemar $p$-values; i.i.d. continuous-score chance Group Score 16.67% [26]).

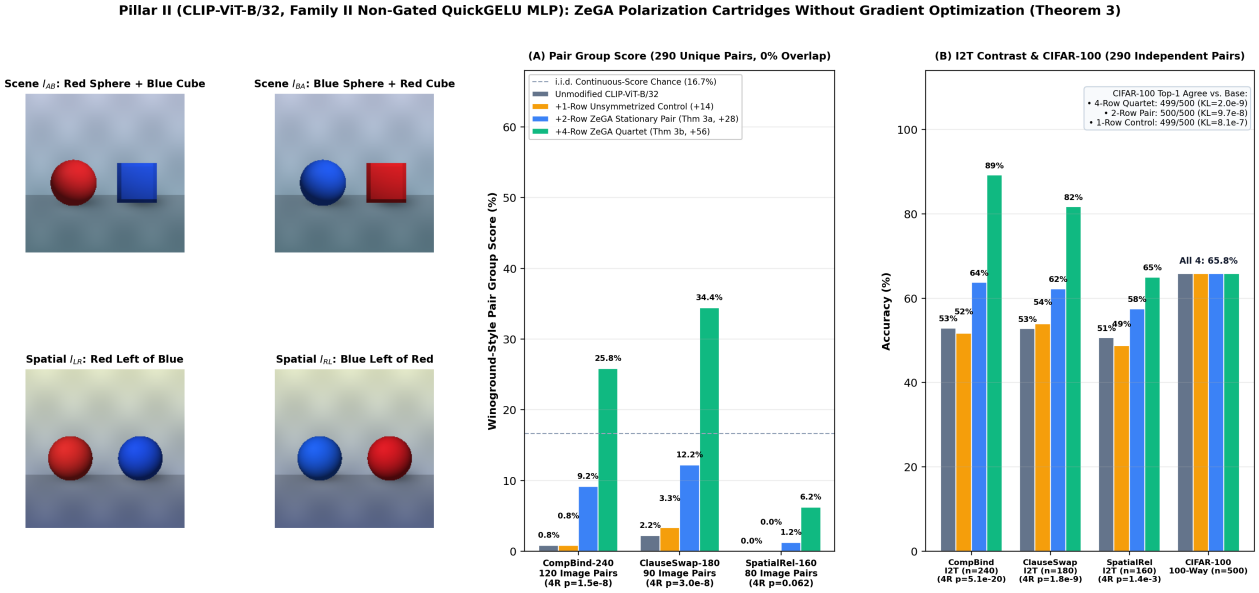



Figure 1: Pillar II (`CLIP-ViT-B/32`, Family II `QuickGELU`): Winoground-style Pair Group Score across 290 unique image pairs (0% template overlap) and `CIFAR-100` preservation (**Theorem 3**).

Table 3: Controlled ablation of **Theorem 3** on `CLIP-ViT-B/32` Layer 10 (290 unique image pairs, 0% template overlap + 700 natural photos).

| Condition | Rows | CompBind Grp ($p$) / I2T | ClauseSwap Grp ($p$) / I2T | SpatialRel Grp ($p$) / I2T | CIFAR-100 Agree / KL | CelebA Probe / Cos |
|---|---|---|---|---|---|---|
| Base `CLIP-ViT-B/32` | +0 | 0.83% / 52.92% | 2.22% / 52.78% | 0.00% / 50.62% | Base (65.8%) / 0 | Base / 1.0000000 |
| +1-Row Control | +14 | 0.83% (1.0) / 51.67% | 3.33% (1.0) / 53.89% | 0.00% (1.0) / 48.75% | 499/500 / 8.08e-7 | 200/200 / 1.0000000 |
| +2-Row Pair (Thm 3a) | +28 | 9.17% (6.3e-3) / 63.75% | 12.22% (0.022) / 62.22% | 1.25% (1.0) / 57.50% | **500/500** / 9.74e-8 | 200/200 / 1.0000000 |
| **+4-Row Quartet (Thm 3b)** | **+56** | **25.83%** (**1.5e-8**) / **89.2%** | **34.44%** (**3.0e-8**) / **81.7%** | **6.25%** (0.062) / **65.0%**$^*$ | 499/500 / **1.99e-9** | **200/200** / **1.0000000** |

# 6 Pillar III: Spatiotemporal Patch-Gated Diffusion on `Juggernaut XL (SDXL)`

On `juggernautXL_ragnarokBy.safetensors` (7,105,350,162 bytes, SHA-256 `dd08fa32f98d05a2443ca1419e46df1575a0811f6e3b246d9dd47ff20f5eb66a`, SDXL-derived [21] 2.57B UNet; 45 tensor payloads across 30 projections verified via SHA-256, 7.16 ms swap), Ridge/SVD weights (+8/5,120 channels across 15 `GEGLU` blocks, 906.84 KiB) were compiled from 4 single-object vessel pairs and 5 anchors (in-sample patch `ROC-AUC` = 0.9976–1.0000), with scalar offset/gain tuned on development scene `puppy_teacup_s104`. Table 4 reports all 8 multi-object scenes, $n = 7$ excluding exact development scene `s104`, and $n = 6$ across the 3 other prompt families (excluding `s104,s208`), using a binary 55%-quantile off-target proxy mask, center-separated crops (IoU = 0.0318), and signed mean positive-CFG-branch `GELU` gate maps (without value product). On `bunny_vase_s519`, the output image exhibits an unconverted white vase (0.03% sapphire), visible sapphire ornament on the rabbit (1.61% naturalness), and low gate-map max (0.2604 vs. 0.5528 on `s623`); we hypothesize shared bright cream/white surface statistics under-drove the fixed threshold.

# 7 Limitations & Conclusion

Corollary 1 bounds single-block MLP residuals rather than global downstream attention/VAE propagation (32.31 dB anchor PSNR). In Family II, Theorem 3b doubles row count (+56 vs. +28),

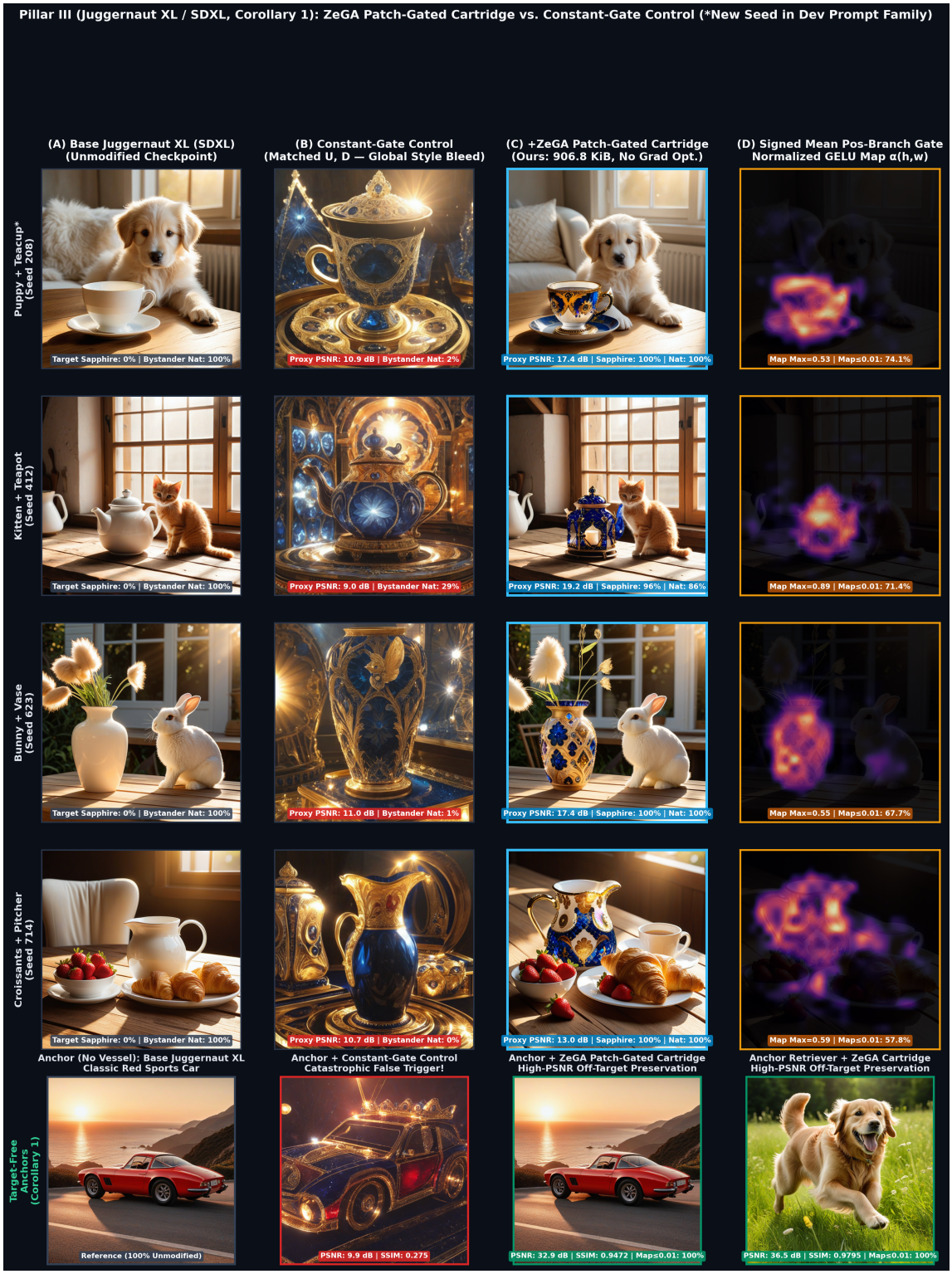



Figure 2: Pillar III (`Juggernaut XL / SDXL`): Base, Constant-Gate Control, `+ZeGA Patch-Gated Cartridge`, and signed mean positive-CFG-branch `GELU` gate maps (`*s208` is a new seed in the development prompt family).

Table 4: Paired evaluation on `Juggernaut XL / SDXL` (768 × 768, 18 Euler steps, CFG = 6.0).

| Subset & Condition | Target Sapphire | Bystander Nat. | Proxy / Anchor PSNR | SSIM | `Map <= 0.01` |
|---|---|---|---|---|---|
| `MultiObj` All ($n = 8$): Base / Const.-Gate Ctrl | 0.16%/99.99% | 100.0%/4.11% | ∞/10.35 dB | 1.000/0.3164 | — / 0.00% |
| `MultiObj` **All** ($n = 8$)**:** `+ZeGA Patch-Gated` | **85.94**% | **85.91**% | **16.36** dB | **0.6741** | **67.84**% |
| `MultiObj` **Excl. Dev Scene** `s104` ($n = 7$)**:** `+ZeGA` | **83.99**% | **83.90**% (4.70% Ctrl) | **16.32** dB (10.32 Ctrl) | **0.6706** | **66.47**% |
| `MultiObj` **Other** 3 **Families** ($n = 6$)**:** `+ZeGA` | **81.33**% | **81.21**% (5.09% Ctrl) | **16.15** dB (10.22 Ctrl) | **0.6615** | **65.19**% |
| `AnchorFree-4` ($n = 4$): Const.-Gate Ctrl | — | — | 10.22 dB | 0.2297 | 0.00% |
| `AnchorFree-4` ($n = 4$)**:** `+ZeGA Patch-Gated` | — | — | **32.31** dB | **0.9495** | **100.00**% |

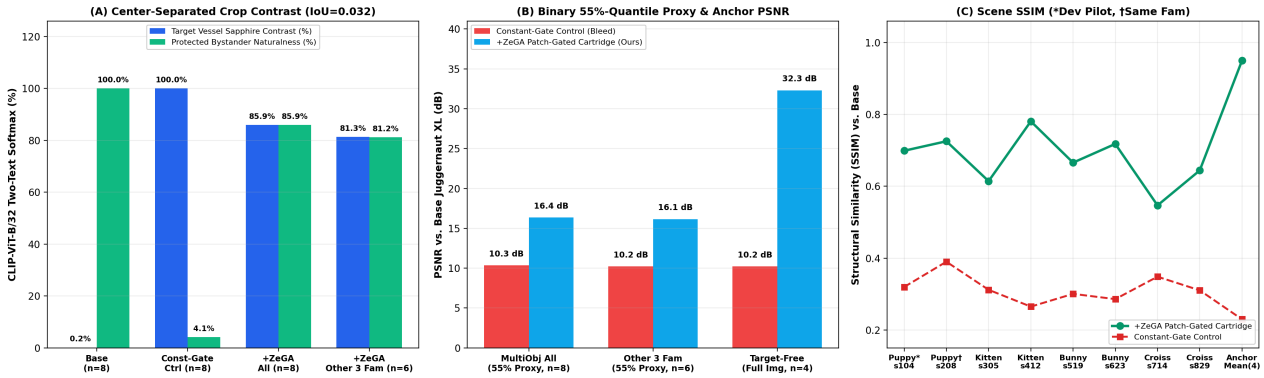



Figure 3: Pillar III quantitative telemetry across all scenes ($n = 8$), the 3 other prompt families ($n = 6$), and target-free anchors ($n = 4$).

`SpatialRel-160` Group Score (6.25% quartet, 1.25% pair; quartet I2T 65.0%*, $p = 0.0014$) remains below the 16.67% i.i.d. reference, and Section 5 evaluates synthetic 3D attribute/relation vocabularies (0% template overlap) rather than open-vocabulary natural benchmarks such as *Winoground* [26] or *SugarCrepe* [10]. In Section 3.4, `argmax` preservation under `RMSNorm` is an *idealized* direct-readout condition ($\mathbf{A}k_{\text{null}} = \mathbf{0}$), whereas our rank-$\leq 64$ unembedding sketch requires empirical end-to-end auditing (`Anchor-40` $\overline{D}_{\text{KL}} = 5.66 \times 10^{-5}$). Audio, video, trained diffusion adapter comparisons, and Family III convolutions remain future work. Overall, **ZeGA** demonstrates reversible, sub-megabyte (337–961 KiB), gradient-free gated adaptation across language, vision–language, and SDXL image generation via a single 272-line native `nn.Linear` loader.

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